

Complex Fragment Coupling via *O*-Mesyl Hydroxylamines: Direct Conversion of Primary Amines to Electrophilic Aminating Agents

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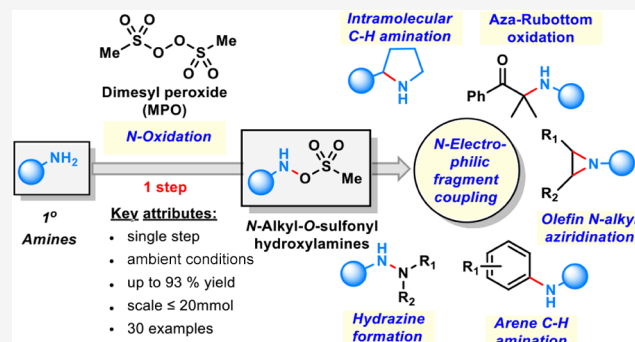


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ABSTRACT: Nitrogen-containing molecules are ubiquitous in pharmaceuticals, natural products, ligands, and functional materials, sustaining continued interest in efficient C–N bond-forming strategies. Although electrophilic amination has emerged as a powerful complement to classical nucleophilic substitution and cross-coupling methods, the transfer of structurally complex *NH*-alkyl groups remains limited by the lack of general access to suitably reactive electrophilic aminating reagents. Herein, we report the direct single-step oxidation of primary amines to *N*-alkyl-*O*-mesyl hydroxylamines using dimesyl peroxide (MPO). This transformation achieves an umpolung of otherwise nucleophilic amines and provides rapid access to synthetically versatile electrophilic nitrogen building blocks under mild conditions. A broad substrate scope enables the preparation of structurally complex and chiral *N*-alkyl-*O*-mesyl hydroxylamines, including derivatives of amino acids, natural products, and pharmaceuticals. These reagents were directly deployed in established electrophilic amination manifolds, including arene C–H amination, olefin aziridination, and aza-Rubottom oxidation, enabling the transfer of structurally complex and chiral amine fragments. Collectively, this work establishes a general strategy for converting readily available primary amines into electrophilic nitrogen fragments for efficient C–N bond formation and *N*-electrophilic fragment coupling.



INTRODUCTION

Amines and, more broadly, nitrogen-containing functional groups are prevalent in a wide range of natural products and constitute key structural motifs in many active pharmaceutical ingredients (APIs).^{1–4} Recent studies have found a sharp increase in the utilization of nitrogen in small-molecule drugs as compared to previous decades.^{5,6} As of today, nitrogen is present in approximately 84% of FDA-approved drugs, with about 82% containing at least one nitrogen heterocycle. Among the top 200 best-selling drugs in 2023, an average of 5.1 nitrogen atoms per molecule is observed, approximately 80% of which are incorporated within *N*-heterocyclic structures.^{5,7} Additionally, nitrogen is commonly found in chemical building blocks, ligands, and various agrochemicals.^{8–10} This ubiquity of nitrogen-containing molecules has driven extensive efforts toward the development of efficient C–N bond-forming reactions. While traditional strategies commonly rely on nucleophilic substitution (*S_N*)¹¹ or metal-catalyzed cross-coupling reactions,^{9,12,13} electrophilic aminations have emerged over the past decade as valuable alternatives.¹⁴ These approaches frequently harness the weak N–O bond of hydroxylamine derivatives to transfer nitrogen onto olefins or aromatic moieties.^{15–19}

While many current methods allow the transfer of an unprotected *NH*₂ moiety, the introduction of *NH*-alkyl groups is significantly more difficult, as they require access to suitably

reactive *NH*-alkyl electrophilic aminating reagents (*R*-*NH*-OSO₂R'). However, these reagents are available only via lengthy multistep syntheses (Figure 1B). Therefore, the direct and chemoselective conversion of readily available primary amines to the corresponding highly reactive *NH*-alkyl-*O*-sulfonylhydroxylamines has been identified as a key challenge. While the direct *N*-oxidation of secondary amines with dibenzoylperoxide (BPO) has been employed to generate *N,N*-dialkyl-*O*-benzoylhydroxylamines,^{20,21} primary amines suffer from competing *N*-benzoylation, resulting in diminished yields of *NH*-alkyl-*O*-benzoylhydroxylamines.

Consequently, only a few methods have been reported for the efficient oxidation of primary amines, and the scope largely remains limited to simple alkylamines without any other functional groups.^{22,23} *NH*-Alkyl-*O*-benzoylhydroxylamines feature two competing electrophilic moieties: the carbonyl group (C=O), which is also present in BPO, and the hydroxylamine nitrogen (N–O), substantially limiting their

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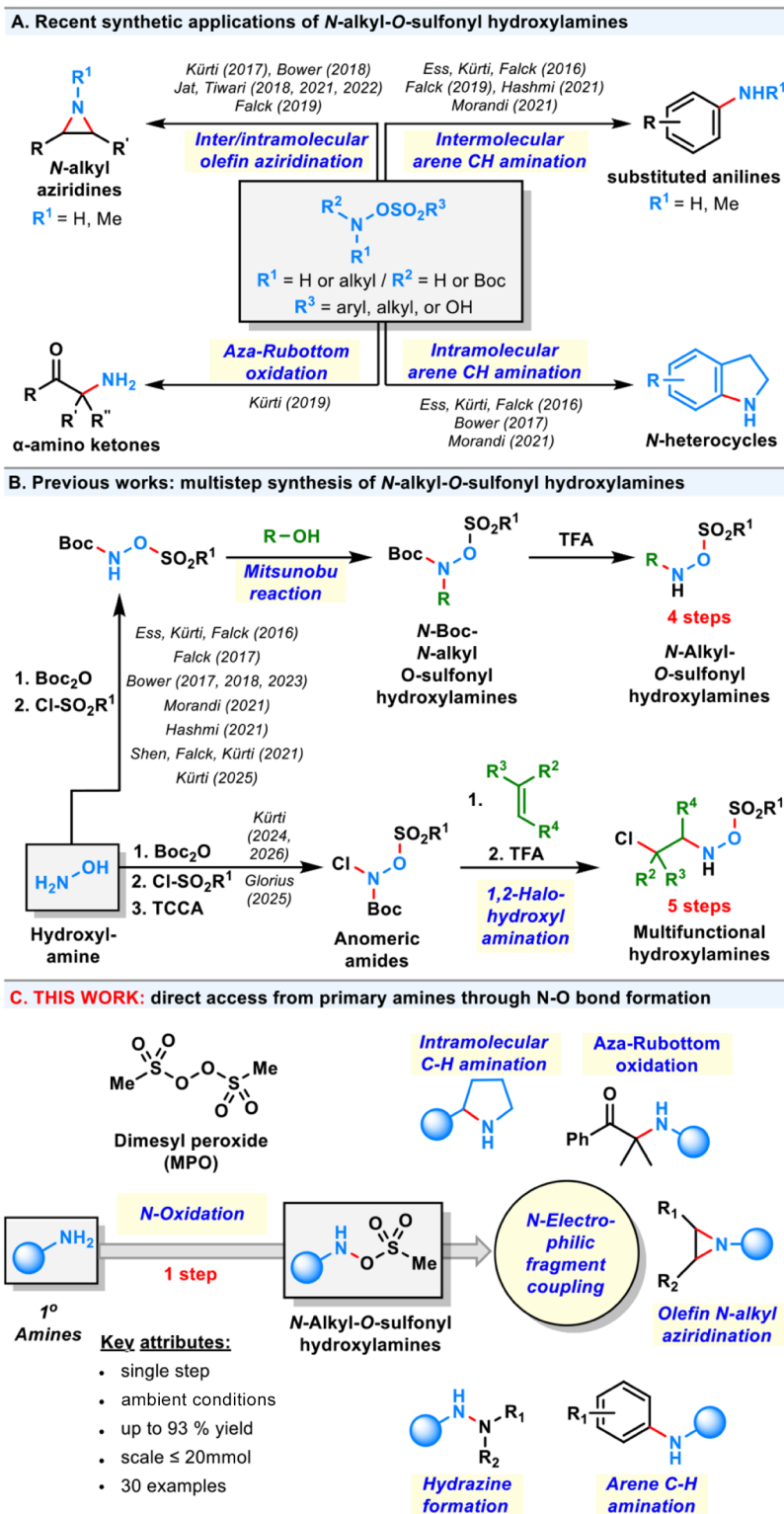


Figure 1. Application and synthesis of *N*-alkyl-*O*-sulfonyl hydroxylamines.

application as electrophilic aminating reagents. Accordingly, most reported electrophilic amination reactions employ secondary *O*-benzoylhydroxylamines, while *NH*-alkyl-*O*-benzoylhydroxylamines remain underrepresented.^{24–26}

When *O*-acylhydroxylamines are directly compared with each other, *NH*-alkyl-*O*-sulfonylhydroxylamines generally exhibit higher reactivity in electrophilic amination reactions than *NH*-alkyl-*O*-benzoylhydroxylamines.

The stronger electron-withdrawing character of the sulfonyl group increases polarization of the N–O bond and facilitates its cleavage, while the resulting sulfonates constitute a superior leaving group compared to carboxylates.²⁷ Thus, multiple *O*-sulfonyl-functionalized aminating agents have been developed over the past two decades (Figure 1A).^{15–18,28–31} As we discussed earlier, the majority of existing methods focus primarily on the transfer of either unprotected NH₂ or simple

NH-alkyl groups (i.e., Me, Et, and *n*-Bu), as more structurally complex electrophilic aminating reagents require multistep syntheses, resulting in overall poor atom economy. Given the lack of efficient methods for the *de novo* construction of the crucial N–O bond, hydroxylamine (HO–NH₂) is often used as the key building block in the syntheses of electrophilic aminating reagents (Figure 1B).

We sought to address this significant gap in the literature by accessing NH-alkyl-*O*-sulfonylhydroxylamines via the direct oxidation of primary amines. Such a transformation in the toolbox of synthetic chemists would substantially expand the scope of electrophilic amination reactions, enabling the facile umpolung of even structurally complex amines and ultimately facilitating the coupling of two structurally complex building blocks.

Recently, Ritter *et al.* reported the use of bis-(methanesulfonyl) peroxide, i.e., dimesyl peroxide (MPO, **1**), in late-stage aromatic C–H oxidations, as well as benzylic oxidations.^{32,33} Intrigued by this reagent, which can be conveniently prepared on the decagram scale by electrolysis of methanesulfonic acid, we considered whether this highly reactive oxidant could enable the direct oxidation of primary amines to the corresponding *O*-activated hydroxylamines. Herein, we report the realization of this concept, enabling the rapid preparation of *O*-sulfonyl hydroxylamines directly from primary amines (Figure 1C) and their application in a large number of electrophilic aminations, including the coupling of structurally complex fragments.

RESULTS AND DISCUSSION

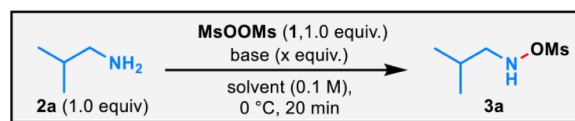
Dimesyl peroxide **1**, an ambient stable and shock-insensitive white microcrystalline solid, was prepared according to the reported electrochemical procedure that involves the anodic oxidation of mesic acid under constant-current conditions in a divided cell.³² With this reagent in hand, we began our investigation of the direct oxidation of primary amines **2** to the corresponding *O*-sulfonyl hydroxylamines (**3**). Adding solid MPO to a solution of isobutylamine **2a** in CDCl₃ rapidly yielded the corresponding *i*-Bu–NH–OMs product (**3a**) while enabling direct and convenient monitoring of the reaction by NMR spectroscopy. The reaction could also be followed by TLC, and both NMR and TLC analysis indicated rapid initial consumption of MPO within minutes at 0 °C.

However, the reaction stalled at approximately 50% conversion, which we attributed to protonation of the amine substrate by the mesic acid that is formed during the reaction. Consistent with a mechanism involving nucleophilic attack of the amine on the peroxide, protonation of the amine likely suppresses its nucleophilicity and thereby inhibits further reaction.

Encouraged by this initial result, we systematically evaluated the reaction parameters. Selected results are summarized in Table 1.

We reasoned that the presence of a base would be required to prevent protonation of the amine substrate and thereby maintain its nucleophilicity. However, initial attempts utilizing organic bases such as triethylamine or Hünig's base (Table 1, entry 6) did not lead to the reaction proceeding past 50% conversion but instead resulted in the formation of a complex mixture, as these bases would also readily react with MPO. Consequently, we focused our attention on inorganic bases. Among those examined, cesium carbonate provided the best results (Table 1, entries 2–5). However, incomplete

Table 1. Optimization of Reaction Conditions



Entry	Base (x equiv)	Solvent	Yield ^a
1	no base	CHCl ₃	25%
2	Cs ₂ CO ₃ (1.0)	CHCl ₃	58%
3	K ₂ CO ₃ (1.0)	CHCl ₃	43%
4	KHCO ₃ (1.0)	CHCl ₃	36%
5	K ₂ HPO ₄ (1.0)	CHCl ₃	35%
6	DIPEA (1.0)	CHCl ₃	-
7	Cs ₂ CO ₃ (1.0)	CH ₃ CN	-
8	Cs ₂ CO ₃ (1.0)	1,2-DCE	41%
9	Cs ₂ CO ₃ (1.0)	DCM	54%
10	Cs ₂ CO ₃ (1.0)	CHCl ₃ ^b	73%/79% ^c
11	K ₂ CO ₃ (1.0)	CHCl ₃ ^b	48%
12	Cs ₂ CO ₃ (2.0)	CHCl ₃ ^b	63%

^aIsolated yields after purification by column. ^b2% v/v H₂O (200 μL) relative to the solvent added. ^cIsolated after 1 M HCl aqueous workup.

consumption of MPO was still observed, and the use of ammonium salts in place of the free amines resulted in little to no conversion to the desired *O*-sulfonyl hydroxylamines. We therefore reasoned that the limited solubility of cesium carbonate in chloroform may be responsible for this lack of deprotonation and screened a range of alternative solvents. Although polar solvents such as acetonitrile, which are known to provide high solubility for cesium carbonate, were evaluated, only chlorinated solvents afforded reliable yields in our hands (Table 1, entries 7–9). Most other solvents failed to deliver isolable amounts of product **3a**, and in several cases, rapid decomposition of MPO was observed. The addition of phase-transfer catalysts such as TBAB or crown ethers also failed to provide reliable yields, often leading to elimination to the corresponding imine or, upon hydrolysis, the corresponding carbonyl compound (i.e., aldehyde).³⁴ These observations suggested that, while the base should be able to neutralize the mesic acid formed during the reaction, it must not promote elimination to the corresponding imine. Finally, the addition of small amounts of water (Table 1, entries 10–12) led to rapid (5–10 min) formation of the desired product, affording *N*-isobutyl *O*-mesyl hydroxylamine **3a** in 73% isolated yield after column chromatography. The NMR yield of this transformation was always higher (~80–85%), suggesting partial decomposition of product **3a** during column chromatography, an effect that was more pronounced for some of the amine substrates. Surprisingly, the *O*-mesyl *N*-alkyl hydroxylamine products (**3**) exhibit significantly reduced basicity at the nitrogen compared to common *O*-benzoyl-*N*-alkylhydroxylamines (R–NH–OBz). This is in line with a recently published report by Jiao *et al.* describing a pK_a value of –0.51 for the triflate salt of the structurally related *O*-tosyl hydroxylamine (NH₂–OTs).³⁵ This diminished basicity allows for an acidic aqueous workup (1 M aqueous HCl) that neutralizes the excess base and removes residual amine starting materials as well as byproducts. As a result, in many cases, the analytically pure products (**3**) are efficiently isolated without the need for additional purification by column chromatography. With the

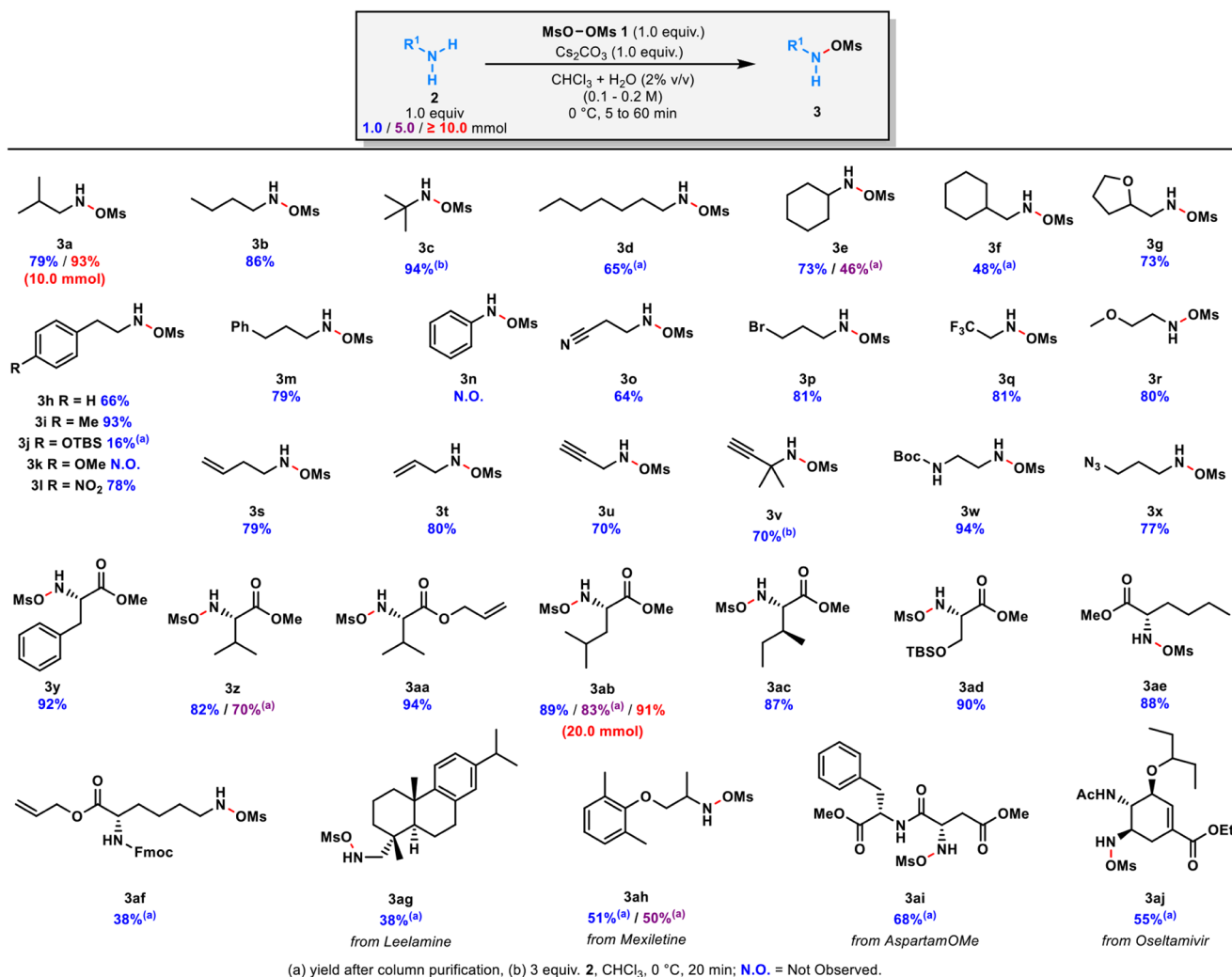


Figure 2. Scope of the direct oxidation of primary amines to *NH*-alkyl-*O*-sulfonyl hydroxylamines.

optimized conditions in hand, we next explored the scope of this transformation (Figure 2).

Simple aliphatic open-chain amines were well-tolerated, affording the corresponding *O*-sulfonyl hydroxylamines in good to excellent isolated yields (**3a–d**). However, primary amines bearing cyclic alkyl groups were not cleanly converted to the products; therefore, chromatographic purification was required to obtain the analytically pure compounds (**3e–f**), resulting in diminished yields. Notably, after aqueous workup, the samples could be directly carried forward in subsequent C–N bond-forming transformations. With this direct approach, overall yields were much improved, indicating that higher amounts of **3** were present in the crude reaction mixtures compared to the samples after chromatographic purification (38% for **3ag** from **2ag**, Figure 2 vs 55% for **20** from **2ag** over 3 steps, Figure 4). Amines bearing pendant aromatic substituents (**3h–m**) were also examined and performed well under the optimized oxidation conditions. However, benzylamine was not compatible, as it underwent spontaneous elimination to the corresponding imine and, upon aqueous workup, formation of benzaldehyde. Furthermore, electron-rich aromatic substituents, such as TBS/Me-protected phenols (**3j–k**), resulted in diminished yields, likely due to subsequent intramolecular amination. Electron-deficient substrates on the other hand, for example, those bearing nitro

groups (i.e., **3l**), were well-tolerated. Anilines (**3n**) were not compatible, as they underwent rapid oxidative decomposition, consistent with the known polymerization of aniline under strongly oxidative conditions.³⁶ A variety of functional groups were well-tolerated, including olefins, halides, nitriles, and ethers, affording the desired products in good to high yields (**3o–x**). Importantly, despite the anticipated high propensity for elimination, amino acid derivatives were well-tolerated and afforded the desired *O*-activated hydroxylamines (**3y–ae**) in excellent isolated yields. Importantly, isoleucine-derived hydroxylamine **3ac** was isolated as a single diastereomer, demonstrating that the oxidation proceeds without epimerization of the stereocenter α to the carbonyl. This single-step oxidation therefore allows the direct use of amines from the chiral pool as powerful electrophilic nitrogen building blocks, effectively achieving an umpolung of the *N*-terminus of amino acids.

However, limitations arise for oxidatively sensitive amino acids (i.e., methionine and tryptophan), which were not compatible under the optimized reaction conditions. While Niciewicz and coworkers successfully employed amino acids in photochemical electrophilic amination reactions,³⁷ isolable amino acid-derived electrophilic aminating reagents have, to the best of our knowledge, not been reported. These resulting RNHOMs reagents can be readily employed across a broad

range of electrophilic amination reactions, greatly expanding the synthetic scope for amino acid-derived electrophilic nitrogen transfer. Finally, the generality of the method was further demonstrated by the successful oxidation of structurally more complex (i.e., functionalized) primary amines derived from natural products and APIs (3af–aj). These substrates were converted to the corresponding *NH*-alkyl-*O*-sulfonyl hydroxylamines in moderate to good yields under standard conditions, highlighting the potential of this transformation for the late-stage modification and fragment coupling of structurally complex building blocks.

The distinctive properties of *NH*-alkyl-*O*-sulfonyl hydroxylamines and the possibility of an unprecedented oxidation pathway prompted us to investigate their electronic structure and reaction mechanism by using density functional theory. Calculations were performed at the UM06–2X/def2-SVP level with a polarizable continuum model (PCM) for chloroform, and all transition structures were validated by intrinsic reaction coordinate (IRC) calculations. We first estimated the pK_a value of *N*-cyclohexyl-*O*-sulfonylhydroxylamine (3e). As anticipated based on the ability to use an acidic workup to purify the hydroxylamines, the *N*-cyclohexyl-*O*-sulfonyl hydroxylammonium has a calculated pK_a value of -6 referenced to the cyclohexylammonium pK_a value of 8 . In addition to the significantly reduced basicity of *N*-cyclohexyl-*O*-sulfonyl hydroxylamine, the OMs group also significantly reduces the nucleophilicity of the nitrogen. This was readily evaluated by comparing the S_N2 barrier heights for cyclohexylamine versus *N*-cyclohexyl-*O*-sulfonyl hydroxylamine for their reaction with methyl chloride as the electrophile. The barrier for *N*-cyclohexyl-*O*-sulfonyl hydroxylamine was calculated to be 10.2 kcal/mol higher than that of cyclohexylamine.

We calculated several reasonable mechanisms for N–O bond construction to form the *NH*-alkyl-*O*-sulfonyl hydroxylamines using 2h and MPO. One simple possible initial step for oxidation of 2h begins with MPO fragmentation to two MeSO_3 radicals, which is a well-known initiation-type event with peroxides. However, we dismissed this bond homolysis pathway because the O–O bond energy in MPO was calculated to be 42.1 kcal/mol, which is consistent with the relative stability of MPO and is much higher than other identified reaction pathways. The O–O bond homolysis of MPO is also unlikely to be initiated by one-electron transfer from the amine, as this process would generate an amine radical cation. This process is estimated to require an enthalpy change of >50 kcal/mol. The only reasonable O–O bond homolysis of MPO occurs if there is electron transfer from Cs_2CO_3 , which requires 17.3 kcal/mol.

Alternative to bond homolysis and electron-transfer mechanisms, we considered nucleophilic amine attack through a S_N2 -at-oxygen mechanism. S_N2 at an oxygen atom, while not well-known, has been proposed in the past.^{20,38–44}

For 2h with MPO, the S_N2 transition structure features a nearly linear arrangement of N–O–O atoms, and the IRC shows a concerted one-step process to form the new NOMs bond and cleave the OOMs bond (Figure 3). The activation enthalpy for this S_N2 transition structure is 25.5 kcal/mol. This S_N2 -at-oxygen barrier can be dramatically reduced if Cs_2CO_3 (or a second amine) acts as a hydrogen-bonding base in the transition state, which enhances nitrogen nucleophilicity. The neutral version of Cs_2CO_3 was modeled with the assumption that the base is relatively insoluble and unlikely to exist as a CsCO_3 anion. The inclusion of Cs_2CO_3 reduces the S_N2 -at-

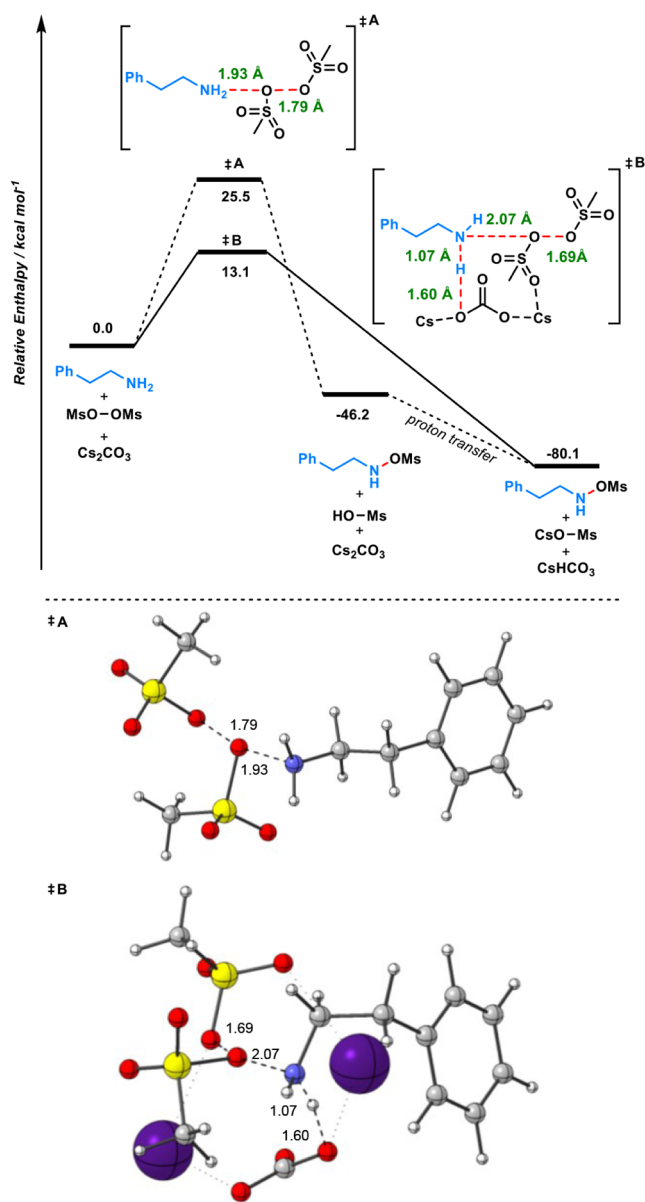


Figure 3. Calculated S_N2 transition structures.

oxygen barrier to only 13.1 kcal/mol. Overall, while it is likely that the S_N2 at oxygen is the lowest-energy mechanism, it is difficult to be definitive about this mechanism since the electron-transfer mechanism with Cs_2CO_3 was calculated to be only a few kcal/mol higher in energy. In order to gain further insight into the reaction mechanism, we performed an additional control experiment using CCl_3Br as a radical scavenger. The reaction proceeded with essentially unchanged yield, further supporting the proposed S_N2 pathway. Additionally, the calculated barriers are consistent with the observed reaction time scales at 0°C .

We next investigated the scalability of the oxidation protocol. A range of RNHOMs reagents was readily prepared on a 5 mmol scale, prompting us to further increase the reaction scale for representative substrates. Thus, 3a and the amino-acid-derived reagent 3ab were prepared on 10 and 20 mmol scales, respectively, affording 1.55 g (93%) and 4.36 g (91%) of the corresponding products and demonstrating the applicability of the protocol on gram and multigram scales.

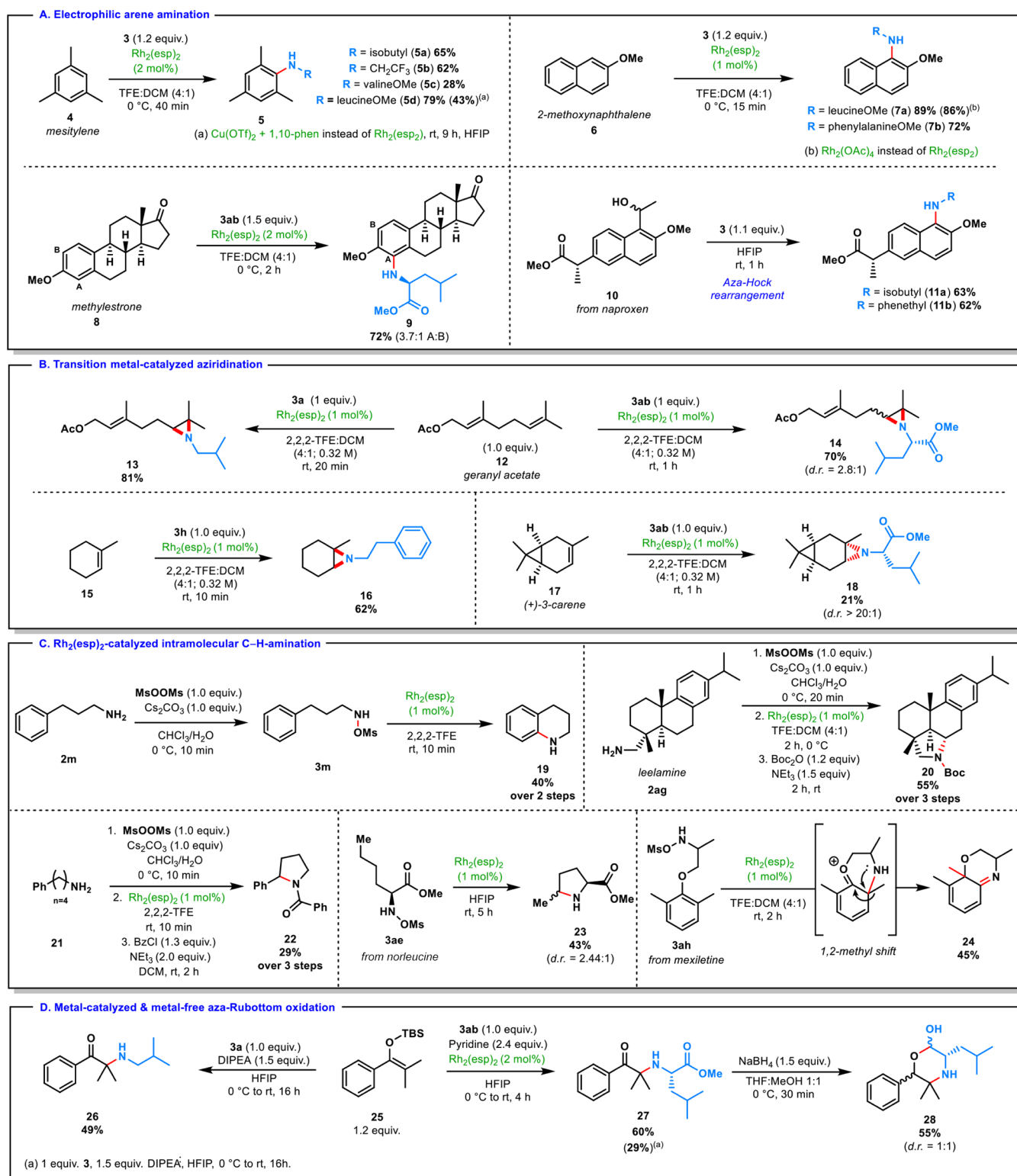


Figure 4. Synthetic application of RNHOMs in N-electrophilic fragment-coupling reactions.

Notably, the transformations performed at larger scales tended to proceed with improved efficiency, and both **3a** and **3ab** were obtained in high purity without the need for column chromatography. Given the exothermic nature of the oxidation, MPO was added portionwise during the scale-up experiments at a rate that maintained the internal reaction temperature below 10 °C. Under these conditions, no operational difficulties were encountered on scales up to 20 mmol,

suggesting that careful control of MPO addition and reaction temperature should be key considerations for further scale-up.

With these O-activated hydroxylamines in hand, we next explored their application in electrophilic amination reactions, deliberately adopting established NHR-transfer protocols.^{15,17,27,31,45–49} In these reactions, commonly used structurally simple aminating reagents (i.e., HOSA, N-alkyl-HOSA, DPH, and N-alkyl-DPH) were directly replaced by

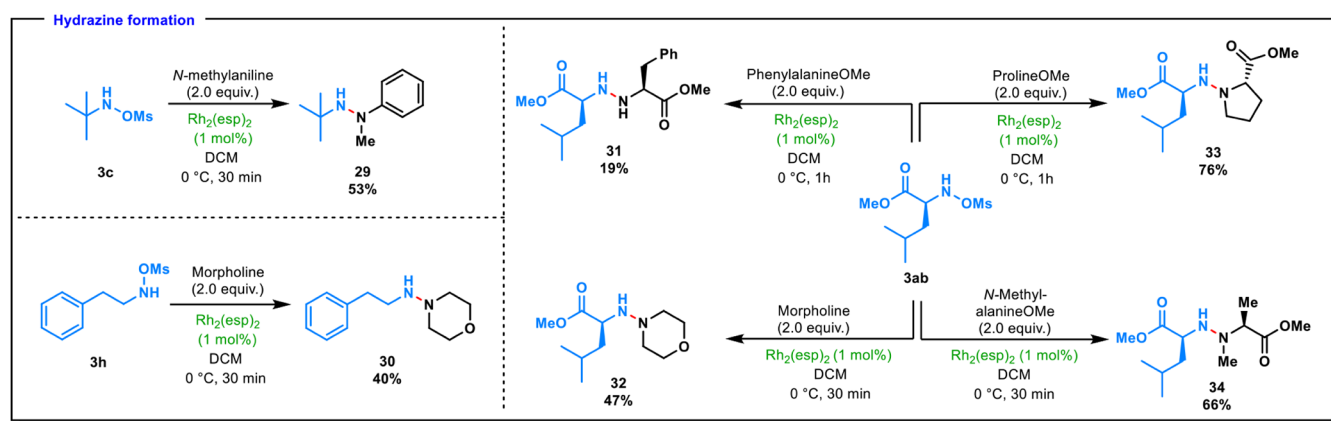


Figure 5. Hydrazine synthesis from two amino building blocks.

more complex RNHOMs derivatives, allowing rapid access to fragment-coupling products with no or only minimal modification of the original conditions. This approach highlights the direct applicability of established electrophilic *N*-transfer protocols without the need for extensive reoptimization.

To demonstrate the direct translation of established electrophilic amination chemistry to complex NHR-transfer reactions, we first examined Rh-catalyst electrophilic aromatic substitutions as previously reported by Falck and Kürti (Figure 4A).^{15,45,50} These protocols proved readily compatible with five different RNHOMs reagents furnishing the corresponding anilines in good to excellent yields (5a–d and 7a–b). Replacing the Du Bois catalyst $\text{Rh}_2(\text{esp})_2$ with the less expensive $\text{Rh}_2(\text{OAc})_4$ for the transformation 6 $\rightarrow$ 7a still resulted in excellent yield (86%), demonstrating that the transformation is not restricted to a single rhodium catalyst. For the arene C–H amination reaction 4 $\rightarrow$ 5d, $\text{Cu}(\text{OTf})_2$ was also a viable catalyst, albeit the product was obtained in a lower yield (43%).⁴⁹ These preliminary results suggest that further optimization and exploration of alternative transition-metal catalysts may provide additional opportunities for electrophilic NHR-transfer reactions. Even structurally complex aryl fragments like methylestrone (8) were well-tolerated, enabling the direct installation of chiral amines, such as leucine-derived motifs, onto complex molecular scaffolds in good yield (72%) and with good regioselectivity (3.7:1). Such chiral fragment-coupling products remain scarcely represented in the literature, highlighting the new opportunities these readily accessible electrophilic aminating agents (RNHOMs) provide for the rapid generation of molecular complexity. Notably, the applicability of RNHOMs reagents extends beyond transition-metal-catalyzed systems. For example, the metal-free aza-Hock rearrangement developed by Hashmi and co-workers⁴⁸ proved readily compatible with the corresponding RNHOMs reagents, enabling the efficient direct arene amination of a naproxen-derived substrate (10, Figure 4A).

The Rh-catalyzed *NH*- and *N*-Me aziridination of unactivated olefins, previously developed by our group and the Falck group,⁴⁷ also proved readily compatible with the structurally more complex RNHOMs reagents, affording the desired *N*-alkylaziridines in good to excellent yields (13 and 14, Figure 4B). Notably, the aziridination of geranyl acetate 12 still proceeded in high yield and with excellent regioselectivity despite the increased steric demand of the RNHOMs reagents compared to previously reported and structurally much simpler

NH_2 -transfer reagents. Remarkably, when an L-leucine-derived RNHOMs reagent (3ab) was employed, aziridination of geranyl acetate proceeded with a diastereomeric ratio of 2.8:1 (14) and in a combined 70% isolated yield. These preliminary results demonstrate the promising synthetic potential of enantiopure RNHOMs reagents to produce structurally complex aziridines with high levels of asymmetric induction.

Intramolecular nitrenoid insertions into $\text{C}(\text{sp}^3)\text{--H}$ as well as $\text{C}(\text{sp}^2)\text{--H}$ have recently been reported by Shibasaki^{51,52} and Falck⁴⁶ utilizing RNHOBz and RNHOTs, respectively, as nitrene precursors. To evaluate whether these transformations are also viable using our novel RNHOMs (3) reagents, the *in situ*-generated *NH*-alkyl-*O*-mesyl hydroxylamines were subjected to the reported reaction conditions (Figure 4C).^{46,51,52} Gratifyingly, both aliphatic $\text{C}(\text{sp}^3)\text{--H}$ and aromatic $\text{C}(\text{sp}^2)\text{--H}$ insertion reactions proceeded smoothly, affording the corresponding cyclic amines in moderate overall yields over two and three steps, respectively (19, 20, and 22). For 20, the initially formed secondary amine was converted directly to the corresponding Boc-carbamate prior to isolation for ease of purification. Remarkably, employing the norleucine-derived hydroxylamine (3ae) afforded the corresponding pyrrolidine 23 in a diastereomeric ratio of 2.44:1. This preliminary result clearly demonstrates the synthetic potential of chiral RNHOMs reagents in diastereoselective $\text{C}(\text{sp}^3)\text{--H}$ amination reactions. Interestingly, exposure of the mexiletine-derived hydroxylamine (3ah) to the Rh-catalyzed $\text{C}(\text{sp}^2)\text{--H}$ amination conditions resulted in the formation of a skeletally rearranged product 24. Based on the observed product structure, we propose an initial dearomative nitrenoid insertion, followed by a rapid 1,2-methyl shift as a plausible reaction pathway.

We next evaluated two structurally complex RNHOMs reagents (3a and 3ab, Figure 4D) in the aza-Rubottom oxidation of silyl enol ethers under the conditions previously reported by our group.¹⁷ This transformation provides direct access to α -aminated carbonyl compounds but has thus far been limited to the transfer of NH_2 groups. To our delight, these RNHOMs reagents reacted readily under both Rh-catalyzed and metal-free conditions, affording the desired α -aminated ketones (26 and 27) in moderate to good yields without further optimization (Figure 4D). Particularly noteworthy were the results obtained with an amino acid-derived RNHOMs reagent (3ab). Following the aza-Rubottom oxidation of a fully substituted silyl enol ether (25), reduction

of the resulting ketone (27) with sodium borohydride triggered spontaneous cyclization between the newly formed secondary alcohol and the pendant amino acid ester moiety, furnishing a highly substituted lactol (28) in a single step. This result demonstrates the rapid assembly of densely functionalized heterocycles from chiral amino acid-derived building blocks, providing straightforward access to enantiomerically enriched morpholinones.

As our group has recently become increasingly interested in N–N bond formation, we sought to explore whether these electrophilic RNHOMs reagents could be directly coupled with readily available classical nitrogen nucleophiles (Figure 5). This strategy would provide a conceptually simple route to unsymmetrical hydrazines from two independently accessible amine fragments. Initial attempts to directly couple *NH*-alkyl-*O*-mesyl hydroxylamines with a variety of amine nucleophiles resulted in elimination to the corresponding imines. We were therefore pleased to find that catalytic amounts of $\text{Rh}_2(\text{esp})_2$ in dichloromethane at 0 °C readily enabled the desired N–N bond formation (29 and 30). Interestingly, the addition of TFE, which proved beneficial in several of the previously discussed transformations, significantly suppressed hydrazine formation under these conditions. Particularly intriguing were coupling reactions involving amino acid-derived reaction partners. Direct coupling of two amino acid fragments would provide access to highly unusual chiral hydrazine derivatives representing a class of compounds that, to the best of our knowledge, remains unexplored. However, coupling of the leucine-derived RNHOMs reagent (3ab) with phenylalanine methyl ester (2y) afforded the desired hydrazine (31) in only 19% yield and with substantial byproduct formation. We hypothesized that oxidation of the initially formed hydrazine (R–NH–NH–R') to the corresponding azo compound (R–N=N–R') might contribute to the diminished yields. To suppress this possible decomposition pathway, we next examined secondary amine nucleophiles, for which analogous azo formation is not possible. Coupling with morpholine resulted in hydrazine formation (32) in higher yield (47%). Substantially improved yields (76% and 66%) of the corresponding hydrazines (33 and 34) were obtained with *L*-proline methyl ester and *N*-methyl *L*-alanine methyl ester as nucleophilic coupling partners. These preliminary results highlight the synthetic potential of RNHOMs reagents to serve as a modular platform for the preparation of densely functionalized hydrazines through the direct coupling of independently accessible enantiomerically pure amine fragments. More broadly, this approach demonstrates how the *N*-umpolung of amino acid-derived amines can enable fundamentally new disconnections for peptide and peptidomimetic synthesis. By reversing the innate polarity of the amino acid *N*-terminus, peptide fragments may be assembled in a manner inaccessible through conventional peptide coupling chemistry, effectively allowing the direction of fragment assembly to be inverted.

CONCLUSION

In conclusion, we have developed a direct and operationally simple oxidation of primary amines to the corresponding *NH*-alkyl-*O*-mesyl hydroxylamines (3) using dimesyl peroxide 1. This transformation provides rapid access to highly versatile electrophilic aminating reagents 3 from abundant amine feedstocks in a single step under mild conditions, effectively achieving an umpolung of otherwise nucleophilic amines. The

resulting RNHOMs reagents (3) were readily prepared from a broad range of substrates, including amino acid derivatives as well as structurally complex natural-product- and pharmaceutical-derived amines.

The versatility of these reagents was demonstrated across a diverse range of electrophilic amination reactions (Figure 4), enabling the direct transfer of structurally complex and chiral amine fragments through established C–H amination, aziridination, and aza-Rubottom-type oxidation protocols. Beyond electrophilic amination chemistry, preliminary studies revealed their utility in N–N bond formation, providing access to densely functionalized hydrazines through direct fragment coupling (Figure 5). We anticipate that this direct amine umpolung strategy will enable new *N*-electrophilic fragment coupling reactions, asymmetric amination methods, efficient N–N bond construction, and previously inaccessible retrosynthetic disconnections for peptide and peptidomimetic synthesis.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.6c14811>.

Detailed experimental procedures, compound characterization data, and copies of NMR spectra (PDF)

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Notes

The authors declare no competing financial interest.

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